

MA378: Tutorial Sheet 5 ANS with some solutions

All questions are from old exam papers; some likely featured on other Tutorial Sheets

Also: Let p_n be the polynomial of degree n that interpolates f at the $n + 1$ points $a = x_0 < x_1 < \dots < x_n = b$. You may assume that, for any $x \in [a, b]$ there is a $\tau \in (a, b)$ such that

$$f(x) - p_n(x) = \frac{f^{(n+1)}(\tau)}{(n+1)!} \pi_{n+1}(x), \quad (1)$$

.....

Q1. (This is Q2, 24/25)

- (a) Let $f(x) = xe^x$. Calculate the Lagrange form of p_2 , the polynomial interpolant to f at the interpolation points $\{x_0, x_1, x_2\} = \{0, 1, 2\}$.

Answer: The Lagrange polynomials are

$$L_0(x) = \frac{1}{2}(x-1)(x-2), \quad L_1(x) = -x(x-2), \quad L_2(x) = \frac{1}{2}x(x-1).$$

Using the L_i , you can find p_2 as

$$p_2(x) = f(x_0)L_0(x) + f(x_1)L_1(x) + f(x_2)L_2(x) = 0L_0(x) + eL_1(x) + 2e^2L_2(x),$$

which can be simplified as $p_2(x) = e(e-1)x^2 - e(e-2)x$.

- (b) Using Equation (1), find an upper bound for $|f(x) - p_2(x)|$, at $x = 1.5$.

Answer: Using (1), the error is given by

$$f(x) - p_2(x) = \frac{f'''(\tau)}{3!}(x-0)(x-1)(x-2),$$

for $\tau \in [0, 2]$. We calculate

$$f'''(x) = 3e^x + xe^x,$$

which is clearly an increasing function on $[0, 2]$, which is maximal at $x = 2$. Thus, $|f'''(\tau)| \leq |f'''(2)| = 5e^2$. Then

$$|f(x) - p_2(x)| \leq \frac{5e^2}{6}|x(x-1)(x-2)|,$$

so that

$$|f(1.5) - p_2(1.5)| \leq \frac{5e^2}{6}|(1.5)(0.5)(-0.5)| = \frac{5e^2}{16} \simeq 2.30908$$

- (c) Let p_n be the polynomial of degree n that interpolates an arbitrary function f at the $n + 1$ equidistant points $-1 = x_0 < x_1 < \dots < x_n = 1$.

Can one expect that $\max_{-1 \leq x \leq 1} |f(x) - p_n(x)| \rightarrow 0$ as $n \rightarrow \infty$? Explain your answer.

Answer: No

This is not the case for all functions. We require that the function's n -th derivative ($f^{(n)}$) is bounded on $-1 \leq x \leq 1$ in the complex plane, otherwise the error will increase with n , e.g. Runge's Example

$$f(x) = \frac{1}{1+x^2},$$

has error that increases with n .

Q2. (This is Q3, 24/25)

- (a) Define the *natural piecewise cubic spline interpolant* to the function f that is continuous on $[a, b]$ at the $N + 1$ knot points $\{x_0, x_1, \dots, x_N\}$.

Answer: The natural cubic spline interpolant to f is the continuous function S such that

- (i) on each interval $[x_{i-1}, x_i]$ for $i = 1, 2, \dots, N$, $S(x) = s_i(x)$ where each s_i is a cubic polynomial, [$S(x)$ is piecewise cubic]
- (ii) $s_i(x_{i-1}) = f(x_{i-1})$ for $i = 1, 2, \dots, N$, [S interpolates f at $N + 1$ points]
- (iii) $s_i(x_i) = f(x_i)$ for $i = 1, 2, \dots, N$, [As in (ii) but also enforces S continuous]
- (iv) $s'_i(x_i) = s'_{i+1}(x_i)$ for $i = 1, 2, \dots, N$, [S has continuous first derivatives on $[x_0, x_N]$]
- (v) $s''_i(x_i) = s''_{i+1}(x_i)$ for $i = 1, 2, \dots, N$, [S has continuous second derivatives on $[x_0, x_N]$]
- (vi) $S''(x_0) = S''(x_N) = 0$. [Natural spline]

- (b) Suppose that S is a natural cubic spline on $[0, 2]$ with

$$S(x) = \begin{cases} 10x + a(1-x)^2 + x^3, & \text{for } 0 \leq x < 1 \\ b(2-x) - c(x+1)^2 + d(x-1)^3, & \text{for } 1 \leq x \leq 2. \end{cases}$$

Find a, b, c and d .

Answer: Using part (a).
The derivatives of S are

$$S'(x) = \begin{cases} 10 - 2a(1-x) + 3x^2, & \text{for } 0 \leq x < 1 \\ -b - 2c(x+1) + 3d(x-1)^2, & \text{for } 1 \leq x \leq 2. \end{cases}$$

and

$$S''(x) = \begin{cases} 2a + 6x, & \text{for } 0 \leq x < 1 \\ -2c + 6d(x-1), & \text{for } 1 \leq x \leq 2. \end{cases}$$

We then apply the conditions of a natural spline (vi).

$$S''(0) = 0 \implies a = 0$$

$$S''(2) = 0 \implies -c + 3d = 0$$

Then any two of the continuity conditions will provide the remaining two equations:

$$\text{Continuity of } S'' \text{ at } x = 1 \implies a + c = -3$$

$$\text{Continuity of } S' \text{ at } x = 1 \implies b + 4c = -13$$

$$\text{Continuity of } S \text{ at } x = 1 \implies b - 4c = 11$$

Finally giving that $a = 0, b = -1, c = -3$ and $d = -1$.

- (c) Suppose that S is the cubic spline interpolant to $f(x) = \ln(x)$ on the $N + 1$ equally spaced points $\{x_0 = 1 < x_1 < \dots < x_N = 4\}$. The error estimate between the function f and the cubic spline interpolant S has an upper bound given by

$$\|f - S\|_\infty := \max_{x_0 \leq x \leq x_N} |f - S| \leq \frac{5h^4}{384} \max_{x_0 \leq x \leq x_N} |f^{(4)}(x)|,$$

where $h = \frac{x_N - x_0}{N}$.

What value of N should one take to ensure that $\|f - S\|_\infty$ is no more than 10^{-6} ?

Answer: The fourth derivative of f is

$$f^{(4)} = -\frac{6}{x^4},$$

so that

$$\max_{1 \leq x \leq 4} |f^{(4)}(x)| = \left| -\frac{6}{(1)^4} \right| = 6,$$

i.e. it is maximal at the minimum value in the range, namely $x = 1$. So for $\|f - S\|_\infty \leq 10^{-6}$

$$\frac{5h^4}{384}(6) \leq 10^{-6} \implies h^4 \leq 0.0000128,$$

and $h = (4 - 1)/N$ so that

$$\left(\frac{3}{N}\right)^4 \leq 0.0000128 \implies N \geq 50.1555228.$$

So one should take $N = 51$ to ensure $\|f - S\|_\infty \leq 10^{-6}$.

Q3. (This is Q4, 24/25) Consider the following differential equation:

$$-u''(x) + u(x) = x \quad \text{on} \quad (0, 1) \quad \text{with} \quad u(0) = u(1) = 0 \quad (2)$$

(a) State the variational formulation of the differential equation (2).

Show that the solution to the variational problem is unique.

Answer: Assume (2) holds and then for any function $v(x)$ we have

$$-u''(x)v(x) + u(x)v(x) = xv(x),$$

which we integrate to find

$$-\int_0^1 u''(x)v(x)dx + \int_0^1 u(x)v(x)dx = \int_0^1 xv(x)dx.$$

Integrating the first term by parts we find

$$-\int_0^1 u''(x)v(x)dx = -u'(x)v(x)\Big|_0^1 + \int_0^1 u'(x)v'(x)dx,$$

which holds for any v , including those where $v(0) = v(1) = 0$, i.e. $v \in H_0^1(0, 1)$. Therefore the variational form is

$$\implies \int_0^1 u'(x)v'(x)dx + \int_0^1 u(x)v(x)dx = \int_0^1 xv(x)dx.$$

Let $\mathcal{A}(u, v) = \int_0^1 u'(x)v'(x) + u(x)v(x)dx = (u', v') + (ru, v)$, where $(\cdot, \cdot)$ is the usual inner product. Suppose there are two solutions u and w then

$$\begin{aligned} \mathcal{A}(u, v) &= (x, v) \quad \text{for all } v \in H_0^1(0, 1), \\ \mathcal{A}(w, v) &= (x, v) \quad \text{for all } v \in H_0^1(0, 1), \\ \implies \mathcal{A}(u - w, v) &= 0 \quad \text{for all } v \in H_0^1(0, 1). \end{aligned}$$

Since u and w are both in $H_0^1(0, 1)$, so too is $u - w$. So we can take $v = u - w$ and

$$\implies \mathcal{A}(u - w, u - w) = 0 \implies u - w = 0.$$

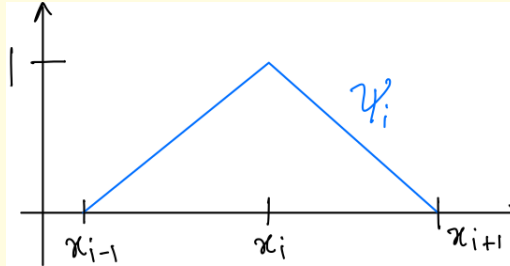
- (b) Suppose you wanted to compute an approximation to (2) using the finite element method (FEM) on the uniform mesh $\{x_0, x_1, \dots, x_n\}$. Denote the set of the usual piecewise linear Galerkin basis (“hat”) functions as $\{\psi_1, \psi_2, \dots, \psi_n\}$.

- i. Give a formula for a typical basis function ψ_i and give a rough sketch of the basis function.

Answer: The basis functions ψ_i are

$$\psi_i(x) = \begin{cases} \frac{x-x_{i-1}}{h} & x_{i-1} \leq x < x_i \\ \frac{x_{i+1}-x}{h} & x_i \leq x \leq x_{i+1} \\ 0 & \text{otherwise} \end{cases}$$

where $h = x_i - x_{i-1}$. A sketch of the basis function ψ_i is below.



- ii. The FEM applied to (2) leads to a linear system of equations, which can be written as the matrix-vector equation $Ax = b$. Give an expression for the entries of A and b in terms of the basis functions ψ_i . (You do not have to derive an explicit formula for these entries.)

Answer: The variational form of (2) is

$$\mathcal{A}(u, v) = (x, v), \quad \text{for all } v \in H_0^1(0, 1)$$

as shown in (a). Using the Galerkin basis functions from (i) (i.e. applying FEM to (2)) we have

$$\mathcal{A}(u_h, \psi_i) = (x, \psi_i) \quad \text{for } i = 1, 2, \dots, N-1, \quad (3)$$

where

$$u_h(x) = \sum_{i=1}^{N-1} \lambda_i \psi_i(x),$$

is the approximation with $N-1$ unknowns λ_i . Thus we can write the system of equations (3) in matrix-vector form with

$$A_{ij} = \mathcal{A}(\psi_i, \psi_j), \quad x_i = \lambda_i, \quad b_i = (x, \psi_i).$$

- iii. Explain why A is symmetric and tridiagonal.

Answer: The entries in A are

$$\begin{aligned} A_{ij} &= \mathcal{A}(\psi_i, \psi_j) = \int_0^1 \psi'_i(x) \psi'_j(x) dx + \int_0^1 \psi_i(x) \psi_j(x) dx \\ &= \int_0^1 \psi'_j(x) \psi'_i(x) dx + \int_0^1 \psi_j(x) \psi_i(x) dx = \mathcal{A}(\psi_j, \psi_i) = A_{ji} \end{aligned}$$

so A is symmetric.

The basis functions $\psi_i(x)$ are only non-zero on $[x_{i-1}, x_{i+1}]$. Therefore if $i > j + 1$ or $j > i + 1$, then $\psi_i(x) \psi_j(x) = 0$ for all x . As above,

$$A_{ij} = \mathcal{A}(\psi_i, \psi_j) = \int_0^1 \psi'_i(x) \psi'_j(x) dx + \int_0^1 \psi_i(x) \psi_j(x) dx.$$

So if $|i - j| > 1$, then $A_{ij} = 0$, i.e. A is tridiagonal.

Q4. (Q2, 23/24)

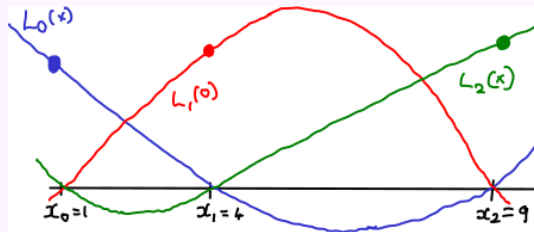
- (a) Let $\{x_0, x_1, x_2\} := \{1, 4, 9\}$ be a set of interpolation points. Write down the formulae for the associated Lagrange Polynomials, $\{L_0, L_1, L_2\}$, and give a rough sketch of them, clearly indicating their values at the interpolation points. (You can make the sketch in your answer book; graph paper is not needed).

Suppose we define $q(x) := 1 - L_0(x) + L_1(x) + L_2(x)$. Show that, in fact, $q(x) \equiv 0$, for all x .

Answer: The formulae are

$$L_0(x) = \frac{1}{24}(x-4)(x-9), \quad L_1(x) = -\frac{1}{5}(x-1)(x-9), \quad L_2(x) = \frac{1}{40}(x-1)(x-4), \quad (4)$$

Sketch:



To show that $q(x) = 0$, one can apply direct calculation. Alternatively, observe that q is a polynomial of degree 2, and $q(1) = q(4) = q(9) = 0$. So $q(x) = 0$ for all x .

- (b) Let $f(x) = x^{3/2}$. Give the Lagrange form of p_2 , the polynomial interpolant to f at $x_0 = 1$, $x_1 = 4$, and $x_2 = 9$.

Using (1), give an upper bound for $|f(7) - p_2(7)|$.

Answer: Using the L_i as given in (4), one can state p_2 as

$$p_2(x) = f(x_0)L_0(x) + f(x_1)L_1(x) + f(x_2)L_2(x) = L_0(x) + 8L_1(x) + 27L_2(x).$$

(This simplifies further as $p_2(x) = 11/60x^2 + 17/12x - 3/5$, but that is not necessary).

For the error estimate, we'll use that

$$f(x) - p_2(x) = \frac{f'''(\tau)}{3!}(x-1)(x-4)(x-9).$$

To find the upper bound, first note that

$$f'''(x) = -\frac{3}{8x^{3/2}}$$

is a negative, increasing function on $[1, 9]$. Hence $|f'''(\tau)| \leq |f'''(1)| = 3/8$. Then

$$|f(7) - p_2(7)| = \frac{3}{8} \frac{1}{6} (6)(3)(2) = 9/4.$$

(Similar to a question on a problem sheet).

Q5. (Q3 23/24)

- (a) Let l be the piecewise linear interpolant to a given function f on $N + 1$ *equally spaced* points in the interval $[a, b]$. Let $h = (b - a)/N$. Use (1) to give an upper bound for $\max_{a \leq x \leq b} |f(x) - l(x)|$, in terms of h and f'' .

Answer: On any interval $[x_i, x_{i+1}]$, l is equivalent to the p_1 Lagrange interpolant to f on $[x_i, x_{i+1}]$, which we denote $l_i(x)$. Therefore, from (1), on that interval

$$f(x) - l_i(x) = \frac{f''(\tau)}{2}(x - x_i)(x - x_{i+1}).$$

Thus,

$$|f(x) - l_i(x)| = \max_{x_i \leq x \leq x_{i+1}} |f''(x)| \frac{1}{2} \max_{x_i \leq x \leq x_{i+1}} |(x - x_i)(x - x_{i+1})|.$$

One can show (in various ways) that $|(x - x_i)(x - x_{i+1})|$ is maximised when $x = (x_i + x_{i+1})/2$. From that, we get

$$|f(x) - l_i(x)| = \max_{x_i \leq x \leq x_{i+1}} |f''(x)| \frac{1}{2} (h/2)(h/2) \leq \frac{h^2}{8} \max_{x_0 \leq x \leq x_N} |f''(x)|.$$

Since the right-hand side is independent of i , we get

$$|f(x) - l(x)| = \frac{h^2}{8} \max_{x_0 \leq x \leq x_N} |f''(x)|. \quad (5)$$

- (b) Let $f(x) = \ln(x)$, $a = 1$, $b = 3$ and $N = 2$. Write down a formula for its piecewise linear interpolant l .

Give an upper bound for $\max_{a \leq x \leq b} |f(x) - l(x)|$.

What value of N would you have to choose to ensure that $\max_{a \leq x \leq b} |f(x) - l(x)| \leq 10^{-4}$?

Answer: In general,

$$l(x) = \begin{cases} \ln(x_0)(x_1 - x) + \ln(x_1)(x - x_0) & x_0 \leq x \leq x_1 \\ \ln(x_1)(x_2 - x) + \ln(x_2)(x - x_1) & x_1 \leq x \leq x_2 \end{cases}$$

For us, $f(x) = \ln(x)$ and $x_0 = 1$, $x_1 = 2$, and $x_2 = 3$. So (and using that $\ln(1) = 0$),

$$l(x) = \begin{cases} \ln(2)(x - 1) & 1 \leq x \leq 2 \\ \ln(2)(3 - x) + \ln(3)(x - 2) & 2 \leq x \leq 3 \end{cases}$$

To bound $|f(x) - \ln(x)|$ on $[1, 3]$ use (5). Since $f''(x) = -1/x^2$, we know that $|f''(x)| \leq 1$ on $[1, 3]$. Also, for this case, $h = 1$. So $|f(x) - \ln(x)| \leq 1/8$ for all $x \in [a, b]$.

To find N such that $|f(x) - \ln(x)| \leq 10^{-4}$ for all $x \in [a, b]$, we need to find h such that

$$\frac{h^2}{8} \max_{x_0 \leq x \leq x_N} |f''(x)| \leq 10^{-4}.$$

Again using that $|f''(x)| \leq 1$ on $[a, b]$, we need h such that $h^2 \leq 8 \times 10^{-4} = 0.02828$. That gives $N \geq (b - a)/h = 70.17$. Rounding up, we see we must take $N = 71$ or larger.

Q6. (Q4, 23/24)

(a) We denote the Newton-Cotes quadrature rule for approximating $\int_{-1}^1 f(x)dx$ as

$$Q_N(f) := \sum_{i=0}^N q_i f(x_i).$$

Show that $q_i = q_{N-i}$ for $i = 0, 1, \dots, N$.

Answer: Several proofs are possible. Here we will use that, by definition $q_i = \int_{-1}^1 L_i(x)dx$, where L_i is the i th Lagrange polynomial for these quadrature points. Since they are evenly spaced on $[-1, 1]$ we know that $x_i = x_{N-1-i}$. That gives that $L_i(x) = L_{N-1-i}(1-x)$. It follows, from a simple change of variables, $y = 1 - x$, that

$$q_i = \int_{-1}^1 L_i(x)dx = - \int_1^{-1} L_{N-1-i}(y)dy = q_{N-1-i}.$$

(This was on a homework assignment).

(b) Consider the following rule for approximating $\int_0^4 f(x)dx$:

$$R(f) = q_0 f(0) + q_1 f(1) + \frac{2}{9} f(3) + f(4).$$

i. Determine the values of q_0 and q_1 so that $R(\cdot)$ has precision 1.

Answer: A method has precision 1 if it is exact for all polynomials of degree 1 or less. If we require that it be exact for $f(x) = 1$, then we get the equation $q_0 + q_1 = 25/9$. If we require that it be exact for $f(x) = x$, then we get the equation $q_1 = 10/3$. From this we determine that $q_0 = -5/9$.

ii. What is the maximum precision of $R(\cdot)$ for the values of q_0 and q_1 that you have determined?

Answer: First note that, since both the integral operator and $R(\cdot)$ are linear, if $R(\cdot)$ is correct for $1, x, x^2, \dots, x^n$ it is exact for all polynomials of degree n or less.

We already know that $R(\cdot)$ has precision of at least 1.

Next we check it for $f(x) = x^2$. For that we get $R(x^2) = 64/3 = \int_0^4 x^2$, so it has precision at least 2.

Then we check it for $f(x) = x^3$. For that we get $R(x^3) = 220/3 \neq 64 = \int_0^4 x^3$, so it does not have precision 3.

Next we check it for that polynomial. Moreover, since the operation is linear, it is in fact exact for all polynomials of

iii. Why is $R(\cdot)$ not a Newton-Cotes rule, in the strictest sense?

Answer: N-C methods use equally spaced quadrature points, which is not the case here. (Unseen, except on last year's exam paper).